Architecture Decision Record: ADR-042

Project: Nexus Core Ledger & Payment Gateway Migration | **Status:** Approved | **Author:** Elena Vance (Lead Solutions Architect) | **Date:** October 14, 2026

1. Context & Problem Statement

The legacy ledger relies on a monolithic DynamoDB NoSQL cluster with custom eventual-consistency resolution logic. This has caused reconciliation discrepancies exceeding $42,000 during high-throughput flash events. The engineering organization requires a deterministic, ACID-compliant ledger storage solution capable of supporting 8,500 financial transactions per second with real-time auditability.

2. Approved Decision & Rationale

We have decided to adopt PostgreSQL 16 with TimescaleDB hypertables for all double-entry transaction ledgers, replacing the DynamoDB store.

> **Documented Rationale:**
> 1. Strict ACID Compliance: Immediate row-level locking and strict serializability are mandatory for cross-border payment settlement.
> 2. Audit Query Performance: TimescaleDB hypertables provide native chunking by timestamp, reducing 90-day forensic audit query execution times from 4 minutes to under 650ms.
> 3. Regulatory Requirements: Compliance standards (FINRA & European Central Bank DORA directives) require verifiable point-in-time state reconstruction.

Lead Owner: Elena Vance (Lead Solutions Architect)

Target Delivery: Sprint 43 (October 24, 2026)

Status: Approved and Implementation In Progress

3. External Dependencies & Blockers

Critical Dependency: Core Banking Gateway API v3.4 release by 3rd-party vendor (FinVance Inc.). Dependency Status: UNRESOLVED. Vendor support ticket #FV-9021 remains open without an SLA commitment or staging endpoint delivery date.

4. Key Management & Security

A decision was recorded to rotate root AWS KMS customer-managed signing keys on a 14-day cycle. (Note: No rationale or compliance justification was provided for this 14-day frequency; standard security policy specifies 90-day rotation).